1. Spaces, Span, Independence

- **Subspace** $W \subseteq V$: $0 \in W$, closed under $+$ and scalar $\cdot$. Intersections are subspaces; unions usually not.
- $\mathrm{span}\{v_1 \dots v_m\} = \{\sum a_i v_i\}$ = smallest subspace containing them.
- **LI**: $\sum a_i v_i = 0 \Rightarrow$ all $a_i = 0$. **LD** $\iff$ some $v_j \in$ span of the others.
- **Basis** = LI + spanning $\iff$ every v has a *unique* representation. $\dim V = \#$basis.
- Any LI list $\leq$ any spanning list. In dim n: n LI $\Rightarrow$ basis; n spanning $\Rightarrow$ basis.
- $\dim(U + W) = \dim U + \dim W - \dim(U \cap W)$.
- **Direct sum** $V = U \oplus W$. Equivalent characterizations: (i) $V = U + W$ *and* $U \cap W = \{0\}$; (ii) every $v \in V$ has a *unique* $v = u + w$; (iii) the only way to write $0 = u + w$ is $u = w = 0$; (iv) $V = U + W$ and $\dim V = \dim U + \dim W$.
- Basis of $U \cup$ basis of W is a basis of $U \oplus W$. Standard instances: $P^2 = P \Rightarrow V = \mathrm{Ker}\, P \oplus \mathrm{Im}\, P$; $V = U \oplus U^{\perp}$ for any subspace U (orthogonal decomposition).

2. Linear Maps

- $T(\alpha u + \beta v) = \alpha Tu + \beta Tv$; $\mathrm{Ker}\, T = \{v : Tv = 0\}$, $\mathrm{Im}\, T = T(V)$.
- T **injective** $\iff$ $\mathrm{Ker}\, T = \{0\}$**.** Surjective $\iff$ $\mathrm{Im}\, T = W$.
- **Rank–nullity:** $\dim V = \dim \mathrm{Ker}\, T + \dim \mathrm{Im}\, T$.
- $T : V \to V$, $\dim V < \infty$: **injective** $\iff$ **surjective** $\iff$ **invertible** (fails in ∞ dim).
- Trick: $T(z_1 \dots z_m) = \sum z_i v_i$. $\{v_i\}$ spans $\iff$ T onto; $\{v_i\}$ LI $\iff$ T 1-1.
- $P^2 = P \Rightarrow V = \mathrm{Ker}\, P \oplus \mathrm{Im}\, P$, $v = (v - Pv) + Pv$.

3. Matrices, Rank, Systems

- $\mathrm{rank}\, A = \#$pivots$= \dim \mathrm{Col}\, A$ = row rank = col rank; $\dim \mathrm{Nul}\, A = n - \mathrm{rank}\, A$.
- $Ax = b$ consistent $\iff$ $\mathrm{rank}\, A = \mathrm{rank}[A|b]$ $\iff$ $b \in \mathrm{Col}\, A$. Solutions $= x_p + \mathrm{Nul}\, A$.
- Basis of $\mathrm{Col}\, A$ = *original* pivot columns. Basis of $\mathrm{Nul}\, A$: set each free var = 1 in turn.
- $\mathrm{rank}\, A = 1 \iff A = cd^{\top}$, i.e. $A_{jk} = c_j d_k$.
- $\mathrm{rank}(AB) \leq \min(\mathrm{rank}\, A, \mathrm{rank}\, B)$; $\mathrm{rank}\, A = \mathrm{rank}\, A^{\top} = \mathrm{rank}(A^{\top} A)$.
- $(AB)^{\top} = B^{\top} A^{\top}$; $(AB)^{-1} = B^{-1} A^{-1}$.

> **Invertible A $(n \times n)$ — all equivalent** $\det A \neq 0$ $\cdot$ rank n $\cdot$ $Ax = 0$ only $x = 0$ $\cdot$ $Ax = b$ unique $\forall b$ $\cdot$ cols LI $\cdot$ cols span $\mathbb{F}^n$ $\cdot$ 0 not an eigenvalue $\cdot$ $A^{\top}$ invertible $\cdot$ n pivots.

Determinant / trace.

- $\det(AB) = \det A \det B$; $\det A^{\top} = \det A$; $\det A^{-1} = 1/\det A$; $\det(cA) = c^n \det A$.
- Row swap flips sign; scaling a row by c scales det by c; adding a multiple of a row: no change.
- **Triangular/diagonal:** $\det = \prod$ diag., eigenvalues = diagonal entries.
- 2×2: $\det = ad - bc$, $A^{-1} = \frac{1}{\det} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$.
- 3×3: $\det A = a_{11}(a_{22}a_{33} - a_{23}a_{32}) - a_{12}(a_{21}a_{33} - a_{23}a_{31}) + a_{13}(a_{21}a_{32} - a_{22}a_{31})$; for $A = \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix}$, $A^{-1} = \frac{1}{\det A} \begin{pmatrix} ei-fh & ch-bi & bf-ce \\ fg-di & ai-cg & cd-af \\ dh-eg & bg-ah & ae-bd \end{pmatrix}$.
- $\mathrm{tr}\, A = \sum a_{ii} = \sum \lambda_i$; $\det A = \prod \lambda_i$; $\mathrm{tr}(AB) = \mathrm{tr}(BA)$.
- **Symmetric** $A = A^{\top}$; **orthogonal** $Q^{\top} Q = I$ ($Q^{-1} = Q^{\top}$, det $= \pm 1$, preserves norms); **idempotent** $P^2 = P$ ($\lambda \in \{0, 1\}$).

4. Eigenvalues & Diagonalization

- $Av = \lambda v$, $v \neq 0$. **Eigenvalue $\iff$ kernel:** λ eigenvalue $\iff$ $\mathrm{Ker}(A - \lambda I) \neq \{0\}$ $\iff$ $A - \lambda I$ singular $\iff$ $\det(A - \lambda I) = 0$.
- So $E_\lambda = \mathrm{Ker}(A - \lambda I)$ and geo. mult. $= \dim \mathrm{Ker}(A - \lambda I) = n - \mathrm{rank}(A - \lambda I)$.
- Special case $\lambda = 0$: $0 \in \sigma(A) \iff \mathrm{Ker}\, A \neq \{0\} \iff A$ *not* invertible. Eigenvectors for $\lambda = 0$ are exactly the nonzero elements of $\mathrm{Ker}\, A$.
- 2×2: $\lambda^2 - (\mathrm{tr}\, A)\lambda + \det A = 0$.
- Distinct $\lambda \Rightarrow$ eigenvectors LI $\Rightarrow$ n distinct $\lambda \Rightarrow$ diagonalizable (*sufficient, not necessary*).
- $1 \leq$ geo. mult. ($\dim E_\lambda$) $\leq$ alg. mult. **Diagonalizable** $\iff$ **geo = alg for every** λ $\iff$ n **LI eigenvectors.** Repeated λ alone decides nothing (I vs $\left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$).
- $X = [v_1 \cdots v_n]$ (eigenvectors in *columns*), $D = \mathrm{diag}(\lambda_i)$ *in the same order.* $AX = XD$, $A = XDX^{-1}$, $A^k = XD^k X^{-1}$.
- **Similarity** $B = S^{-1}AS$: same λ, det, tr, rank, char. poly. $Av = \lambda v \Rightarrow B(S^{-1}v) = \lambda(S^{-1}v)$.
- **Spectral thm (real symmetric):** λ all real; eigenvectors of distinct λ are $\perp$; $A = Q\Lambda Q^{\top}$.
- **Nilpotent:** $A^k = 0$ some $k \Rightarrow$ all $\lambda = 0$; never invertible, never diagonalizable unless $A = 0$. **Neumann:** $(I - N)^{-1} = I + N + \cdots + N^{k-1}$; $(I + N)^{-1} = I - N + \cdots + (-1)^{k-1}N^{k-1}$.
- **Involution:** $T^2 = I$ ($T = T^{-1}$) $\Rightarrow \lambda = \pm 1$ only.
- **Invariant subspace:** $W \subseteq V$ invariant under T iff $T(W) \subseteq W$; eigenspaces E_λ are the canonical example.

5. Quadratic Forms & Definiteness

$Q(x) = x^{\top} Ax$, A symmetric (else use $\frac{1}{2}(A + A^{\top})$).

	eigenvalues	minors
PD	all $\lambda > 0$	leading $D_k > 0$
ND	all $\lambda < 0$	$D_1 < 0, D_2 > 0, \dots$
PSD	all $\lambda \geq 0$	*all* principal minors ≥ 0
NSD	all $\lambda \leq 0$	$(-1)^k \cdot$minor ≥ 0

- **Sylvester's criterion** (definite case only): A sym. is PD $\iff$ $D_1 > 0, \dots, D_n > 0$, where D_k = det of the top-left $k \times k$ block. ND: apply to $-A$, i.e. signs alternate $D_1 < 0, D_2 > 0, \dots$
- **Leading** D_k: only the n top-left blocks. **Principal** minor: pick *any* index set, keep those rows *and* the same columns ($2^n - 1$ of them). Leading $\subset$ principal.

Sylvester with $\geq$ is FALSE. $\mathrm{diag}(0, -1)$: $D_1 = D_2 = 0 \geq 0$ but it is NSD. The principal minor $\{2\} = -1$ catches it. So **PSD requires all principal minors** ≥ 0, not just leading.

- 2×2 shortcut: PD $\iff$ $a > 0$ and $\det > 0$; ND $\iff$ $a < 0$ and $\det > 0$; indefinite $\iff$ $\det < 0$.
- Eigenvalue test never has this trap — use it when λ are cheap. $A^{\top} A$ always PSD; PD $\iff$ A has LI columns.

6. Inner Products, Gram–Schmidt, Projection

- **Euclidean inner product** on $\mathbb{R}^n$: $\langle u, v \rangle = \sum_i u_i v_i = u^{\top} v$, $\|v\| = \sqrt{\langle v, v \rangle}$. Axioms: linear in 1st arg, (conj.-)symmetric, $\langle v, v \rangle > 0$ for $v \neq 0$. Other instances: $\langle p, q \rangle = \int_0^1 pq\, dx$ on $P_2(\mathbb{R})$ ($\int_0^1 x^k = \frac{1}{k+1}$); $\langle X, Y \rangle = \mathbb{E}[XY]$ on random variables.
- $u \perp v \iff \langle u, v \rangle = 0$. $U^{\perp} = \{v : \langle v, u \rangle = 0\ \forall u \in U\}$ is a subspace; $V = U \oplus U^{\perp}$.
- **Cauchy–Schwarz** $|\langle u, v \rangle| \leq \|u\| \|v\|$ (= iff LD). **Triangle** $\|u + v\| \leq \|u\| + \|v\|$. **Reverse** $\big|\|u\| - \|v\|\big| \leq \|u - v\|$.
- $u \perp v \Rightarrow \|u + v\|^2 = \|u\|^2 + \|v\|^2$.
- **Operator norm** $\|A\| = \max_{\|v\|=1} \|Av\|$, so $\|Ah\| \leq \|A\| \|h\|$ (*not* Cauchy–Schwarz: matrix $\times$ vector, not an inner product).
- **Gram–Schmidt** (subtract the projection onto each previous vector): $u_1 = v_1$, $u_k = v_k - \sum_{j<k} \frac{\langle v_k, u_j \rangle}{\langle u_j, u_j \rangle} u_j$, $e_k = u_k / \|u_k\|$. $\mathrm{span}\{u_1 .. u_k\} = \mathrm{span}\{v_1 .. v_k\}$ at every step.
- **Projection** onto U, o.n. basis: $P_U b = \sum_j \langle b, e_j \rangle e_j$. Residual $b - P_U b \perp U$; $P_U b$ unique closest point of U to b.
- Least squares: $A^{\top} A \hat{x} = A^{\top} b$, $P = A(A^{\top} A)^{-1} A^{\top}$.
- **Idempotent** $P^2 = P \Rightarrow \lambda \in \{0, 1\}$. For $P = X(X^{\top} X)^{-1} X^{\top}$ (X: $n \times k$): $\mathrm{tr}\, P = k$; residual $M = I_n - P$: $\mathrm{tr}\, M = n - k$.
- $\langle u, v \rangle = 0 \iff \|u\| \leq \|u + av\|\ \forall a$ — the least-squares FOC.

7. Norms, Sequences, sup/inf

- $\|x\|_\infty \leq \|x\|_2 \leq \sqrt{n}\, \|x\|_\infty$. All norms on $\mathbb{R}^n$ equivalent $\Rightarrow$ same convergent sequences, open sets, continuous functions.
- $x_k \to x$: $\forall \varepsilon > 0\ \exists N : k \geq N \Rightarrow \|x_k - x\| < \varepsilon$. In $\mathbb{R}^n$ $\iff$ coordinatewise.
- Limits unique; convergent $\Rightarrow$ bounded; every subsequence has the same limit.
- $x_k \not\to x$ (negation, used constantly): $\exists \varepsilon > 0$ and a subsequence with $\|x_{k_j} - x\| \geq \varepsilon\ \forall j$.
- **Cauchy**: $\forall \varepsilon\ \exists N : m, k \geq N \Rightarrow \|x_m - x_k\| < \varepsilon$. $\mathbb{R}^n$ complete: Cauchy $\iff$ convergent.
- **Bolzano–Weierstrass**: every bounded sequence in $\mathbb{R}^n$ has a convergent subsequence.
- Monotone + bounded (in $\mathbb{R}$) $\Rightarrow$ convergent.
- $m = \inf S$: (i) $m \leq s\ \forall s \in S$; (ii) m is the *greatest* such. Working form: $\forall \varepsilon > 0\ \exists s \in S$, $s < m + \varepsilon$ (take $\varepsilon = 1/k$ to get $s_k \to m$). Sup mirrors. Inf need not be attained: $\inf(0, 1] = 0 \notin (0, 1]$.
- Completeness axiom: nonempty $S \subseteq \mathbb{R}$ bounded below has $\inf S \in \mathbb{R}$.

8. Open, Closed, Closure

- $B_\varepsilon(x) = \{y : \|y - x\| < \varepsilon\}$. A **open** $\iff$ $\forall x \in A\ \exists \varepsilon > 0$, $B_\varepsilon(x) \subseteq A$.
- A **closed** $\iff$ A^c open $\iff$ ($x_k \in A$, $x_k \to x \Rightarrow x \in A$) $\iff$ $A = \bar{A}$.
- **Interior** $\mathrm{int}\, A$ = largest open set $\subseteq A = \{x : \exists \varepsilon, B_\varepsilon(x) \subseteq A\}$. A **open** $\iff$ $A = \mathrm{int}\, A$.
- **Closure** $\bar{A}$ = smallest closed set $\supseteq A = A \cup A' = \{x : \forall \varepsilon,\ B_\varepsilon(x) \cap A \neq \emptyset\}$ = all limits of sequences in A.
- **Boundary** $\partial A = \bar{A} \setminus \mathrm{int}\, A = \{x : \forall \varepsilon,\ B_\varepsilon(x)$ meets both A and $A^c\}$. Always closed. $\bar{A} = A \cup \partial A$.
- **Limit (accumulation) point** $x \in A'$: every $B_\varepsilon(x)$ contains a point of A other than x. **Isolated**: some $B_\varepsilon(x) \cap A = \{x\}$.
- **Set algebra:** $A \cup B = B \cup A$, $A \cap B = B \cap A$ (comm.); $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$ (distrib.). **De Morgan:** $(A \cup B)^c = A^c \cap B^c$, $(A \cap B)^c = A^c \cup B^c$; general $\left(\bigcup_i A_i\right)^c = \bigcap_i A_i^c$.
- $\overline{A^c} = (\mathrm{int}\, A)^c$; $\mathrm{int}(A^c) = (\bar{A})^c$ (De Morgan for topology).
- Arbitrary $\bigcup$ of open is open; finite $\bigcap$ of open is open (dually for closed). $\emptyset, \mathbb{R}^n$ are both open and closed.
- Infinite $\bigcap$ of open need not be open: $\bigcap_k (-\frac{1}{k}, \frac{1}{k}) = \{0\}$.
- **Bounded**: $\exists M$ with $\|x\| \leq M\ \forall x \in A$. **Dense**: $\bar{A} = \mathbb{R}^n$ (e.g. $\mathbb{Q}$ in $\mathbb{R}$).

9. Compactness

- **Compact** (Heine–Borel in $\mathbb{R}^n$): closed *and* bounded $\iff$ every open cover has a finite subcover $\iff$ every sequence has a subsequence converging *in* the set.
- Closed subset of compact is compact (bounded from K; limit retained by closedness of F).
- Nested nonempty compacts $K_1 \supseteq K_2 \supseteq \cdots \Rightarrow \bigcap K_k \neq \emptyset$.
- Compact $\Rightarrow$ sup and inf are attained (belong to the set).
- **Separating Hyperplane:** disjoint nonempty convex $C_1, C_2 \subseteq \mathbb{R}^n$, one compact and the other closed $\Rightarrow \exists p \neq 0, c \in \mathbb{R}$ with $p^{\top} x \leq c \leq p^{\top} y\ \forall x \in C_1, y \in C_2$.

10. Limits & Continuity

- $\lim_{x \to a} f(x) = L$: $\forall \varepsilon > 0\ \exists \delta > 0 : 0 < \|x - a\| < \delta \Rightarrow \|f(x) - L\| < \varepsilon$.
- f **continuous at** a: $\forall \varepsilon\ \exists \delta : \|x - a\| < \delta \Rightarrow \|f(x) - f(a)\| < \varepsilon \iff x_k \to a \Rightarrow f(x_k) \to f(a)$.
- **Global:** f continuous $\iff$ f^{-1}(open) open $\iff$ f^{-1}(closed) closed. Preimages, not images.
- Corollaries: $\{f > c\}, \{f < c\}$ open; $\{f \geq c\}, \{f \leq c\}, \{f = c\}$ closed.
- Sums, products, quotients (nonzero denom.), compositions of continuous are continuous.
- **Weierstrass (EVT):** K compact, f cont. $\Rightarrow f(K)$ compact $\Rightarrow$ ($m = 1$) max and min attained.
- **IVT** ($n = m = 1$): f cont. on $[a, b]$, y between $f(a), f(b) \Rightarrow \exists c$ with $f(c) = y$.
- Continuous bijection on compact $K \Rightarrow f^{-1}$ continuous.
- **Uniform continuity**: δ independent of x. Continuous on compact $\Rightarrow$ uniformly continuous. **Lipschitz**: $\|f(x) - f(y)\| \leq L\|x - y\| \Rightarrow$ unif. cont.
- $d(x, A) = \inf_{a \in A} \|x - a\|$ is 1-Lipschitz; $d(x, A) = 0 \iff x \in \bar{A}$.

11. One-Variable Calculus

- $f'(a) = \lim_{h \to 0} \frac{f(a+h) - f(a)}{h}$. Differentiable $\Rightarrow$ continuous (converse false: $|x|$ at 0).
- **Product:** $(fg)' = f'g + fg'$. **Quotient:** $\left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2}$. **Chain:** $(f \circ g)'(x) = f'(g(x))g'(x)$.
- **Inverse:** $(f^{-1})'(y) = \dfrac{1}{f'(f^{-1}(y))}$, valid when $f' \neq 0$ there.
- **Common Derivatives:**

– $x^a : ax^{a-1}$	– $\log_a x : 1/(x \ln a)$
– $e^x : e^x$	– $\sin x : \cos x$
– $a^x : a^x \ln a$	– $\cos x : -\sin x$
– $\ln x : 1/x$	– $\tan x : sec^2 x$

- **Rolle:** f cont. $[a, b]$, diff. (a, b), $f(a) = f(b) \Rightarrow \exists c \in (a, b)$, $f'(c) = 0$.
- **MVT:** same hypotheses $\Rightarrow \exists c \in (a, b)$ with $f'(c) = \dfrac{f(b) - f(a)}{b - a}$.
- Consequences: $f' \equiv 0 \Rightarrow f$ const; $f' > 0 \Rightarrow$ strictly increasing; $|f'| \leq L \Rightarrow L$-Lipschitz.
- **Taylor (1D, Lagrange remainder):** $f(a + h) = \sum_{j=0}^{n} \frac{f^{(j)}(a)}{j!} h^j + \frac{f^{(n+1)}(a + \theta h)}{(n+1)!} h^{n+1}$, $\theta \in (0, 1)$.
- **L'Hôpital:** $\frac{0}{0}$ or $\frac{\infty}{\infty} \Rightarrow \lim \frac{f}{g} = \lim \frac{f'}{g'}$ when the latter exists.
- **FTC:** $\frac{d}{dx} \int_a^x f = f(x)$; $\int_a^b f' = f(b) - f(a)$. **Leibniz:** $\frac{d}{dx} \int_{a(x)}^{b(x)} f(x, t) dt = f(x, b)b' - f(x, a)a' + \int_a^b f_x\, dt$.
- Integration by parts $\int u\, dv = uv - \int v\, du$; substitution $\int f(g)g' = \int f(u) du$.

12. Differentiation in $\mathbb{R}^n$

- **Definition:** $f(x + h) = f(x) + Df(x)h + r(h)$, $\|r(h)\| / \|h\| \to 0$. Df **unique** (put $h = tv$, divide by t, $t \to 0$; then $(A - B)v = 0\ \forall v$).
- Read it as: f *is affine to first order near* x; $Df(x)h$ is the predicted change, $r(h)$ the leftover.

> **Jacobian** $(m \times n)$ $Df(x) = \begin{pmatrix} \partial_1 f_1 & \cdots & \partial_n f_1 \\ \vdots & & \vdots \\ \partial_1 f_m & \cdots & \partial_n f_m \end{pmatrix}$, entry $(i, j) = \dfrac{\partial f_i}{\partial x_j}$

> **Hessian** $(n \times n)$ $H_f(x) = D(\nabla f)(x) = \begin{pmatrix} f_{11} & \cdots & f_{1n} \\ \vdots & & \vdots \\ f_{n1} & \cdots & f_{nn} \end{pmatrix}$, $f_{ij} = \dfrac{\partial^2 f}{\partial x_i \partial x_j}$. **Symmetric if** $f \in C^2$

- **Directional:** $D_v f(x) = \lim_{t\to 0} \frac{f(x+tv)-f(x)}{t}$. If differentiable, $D_v f = Df(x)v$ (*linear in* v). Partials $= D_{e_j} f$.
- **Max Rate Incr Directional:** $\max_{|v|=1} D_v f(x) = |\nabla f(x)|$, attained at $v = \nabla f(x)/|\nabla f(x)|$
- **Directional 2nd Der:** $D_v^2 f(x) = v^\top H_f(x) v$
- Differentiable $\Rightarrow$ continuous. Partials exist $\not\Rightarrow$ continuous.
- $C^1 \Rightarrow$ **differentiable**: split the increment one variable at a time, MVT on each piece $\Rightarrow f_x(\xi, b+k)h + f_y(a,\eta)k$; subtract $f_x(a,b)h + f_y(a,b)k$; bound by $(|\Delta f_x| + |\Delta f_y|)\sqrt{h^2+k^2}$; continuity kills each Δ.
- **Chain rule:** $D(g \circ f)(x) = Dg(f(x))\, Df(x)$, sizes $(p\times m)(m \times n) = p \times n$. Order matters; outer evaluated *at* $f(x)$. Inner Jacobian has **one column per independent variable**.
- Scalar form: $\dfrac{\partial (g\circ f)}{\partial x_j} = \sum_i \dfrac{\partial g}{\partial y_i}\dfrac{\partial f_i}{\partial x_j}$. If x enters twice, both paths contribute (total vs partial derivative).
- $\nabla(a^\top x) = a$; $D(Ax) = A$; $\nabla(x^\top Ax) = (A + A^\top)x = 2Ax$ if A sym.; $H(x^\top Ax) = 2A$.
- ∇f points in the direction of steepest ascent; $\nabla f \perp$ level set $\{f = c\}$.

13. Restrict to a Line

$g(t) = f(a + t(b-a))$, $t \in [0,1]$: *multivariate* $\to$ *segment* $\to$ *one-variable calculus.*

- $g'(t) = Df(a+t(b-a))(b-a)$.
- **MVT:** f real-valued $\Rightarrow \exists c$ on the segment with $f(b) - f(a) = Df(c)(b-a)$. Fails for $m \geq 2$ (each component gets its own c); use $|f(b)-f(a)| \leq \sup_{z\in[a,b]} \|Df(z)\| \|b-a\|$.
- U convex, $Df \equiv 0 \Rightarrow f$ constant. $H_f \equiv 0 \Rightarrow D(\nabla f) \equiv 0 \Rightarrow \nabla f \equiv a \Rightarrow f(x) = a^\top x + b$ (apply the previous line to $g(x) = f(x) - a^\top x$).
- **Second-order Taylor approximation** (the working form): for h small,

$$f(x+h) \;\approx\; f(x) + \nabla f(x)^\top h + \tfrac{1}{2}\, h^\top H_f(x)\, h,$$

  with error $o(\|h\|^2)$. First-order: $f(x+h) \approx f(x) + \nabla f(x)^\top h$ (the tangent plane).
- Everything is evaluated *at* x: gradient $\nabla f(x)$ (a vector, paired with h) and Hessian $H_f(x)$ (a matrix, sandwiched as $h^\top Hh$). The second-order term is a *scalar quadratic form* because f is scalar-valued.
- At a critical point $\nabla f(x^*) = 0$, so $f(x^*+h) - f(x^*) \approx \frac{1}{2} h^\top H_f h$: the sign of the quadratic form *is* the local min/max verdict. That is why definiteness is the SOC.

14. Inverse & Implicit Function Theorems

Inverse FT $f : \mathbb{R}^n \to \mathbb{R}^n$ C^1 near x_0, $\det Df(x_0) \neq 0$ $\Rightarrow \exists$ open $U \ni x_0$, $V \ni f(x_0)$ with $f : U \to V$ a C^1 bijection and

$$D(f^{-1})(f(x_0)) = [Df(x_0)]^{-1}.$$

Local only; sufficient, not necessary.

- $F(x,y) = (e^x \cos y, e^x \sin y)$: $\det DF = e^{2x} \neq 0$ everywhere, yet $F(x, y+2\pi) = F(x,y)$ — locally 1-1, not globally. IFT is local.
- $f(x) = x^3$: $f'(0) = 0$ but globally 1-1; $f^{-1} = x^{1/3}$ *not* differentiable at 0. Singular Df rules out a *differentiable* inverse, not injectivity.
- **Steps:** (1) confirm $f \in C^1$ near x_0; (2) compute $Df(x_0)$, check $\det \neq 0$; (3) conclude local C^1 bijection $U \to V$; (4) invert the Jacobian for $D(f^{-1})(f(x_0))$; (5) local only — check global injectivity separately (e.g. periodicity).

Implicit FT $F : \mathbb{R}^n \times \mathbb{R}^m \to \mathbb{R}^m$, $F(x_0, y_0) = 0$, $F \in C^1$, $D_y F(x_0, y_0)$ *invertible* $\Rightarrow$ locally a unique C^1 map $y = g(x)$ with $g(x_0) = y_0$ and

$$Dg(x) = -[D_y F]^{-1} D_x F.$$

Scalar case $F(x,y,z) = 0$, $F_z \neq 0$: $g_x = -F_x/F_z$, $g_y = -F_y/F_z$.

- Derive by chain rule, not memory: $F(x, y, g(x,y)) = 0 \Rightarrow F_x + F_z g_x = 0$.
- $GL_n = \det^{-1}(\mathbb{R} \setminus \{0\})$ is open $\Rightarrow Df(x^*)$ invertible $\Rightarrow Df(x)$ invertible nearby.
- **Steps:** (1) confirm $F(x_0, y_0) = 0$; (2) compute $D_y F$, check $\det \neq 0$ (scalar: $F_y \neq 0$); (3) conclude unique C^1 $y = g(x)$ near x_0; (4) chain rule on $F(x, g(x)) \equiv 0 \Rightarrow Dg = -[D_y F]^{-1} D_x F$; (5) evaluate at (x_0, y_0) for the numeric derivative.
- **Step4 ext:**

$$\frac{d}{dx}\left[F(x, g(x))\right] = D_x F(x,y) + D_y F(x,y) \cdot Dg(x) = \mathbf{0}$$

;

$$Dg(x) = -[D_y F(x,y)]^{-1} D_x F(x,y)$$

- **Scalar case:**

$$F_x + F_y \frac{dy}{dx} = 0$$

;

$$\frac{dy}{dx} = -\frac{F_x}{F_y}$$

15. Unconstrained Optimization

- **FOC:** interior local extremum + differentiable $\Rightarrow \nabla f(x^*) = 0$.
- **SOC:** $H_f(x^*)$ PD $\Rightarrow$ strict local min; ND $\Rightarrow$ strict local max; indefinite $\Rightarrow$ saddle; semidefinite $\Rightarrow$ inconclusive.
- **Convexity:** f convex on convex $U \iff H_f$ PSD $\iff f(y) \geq f(x) + \nabla f(x)^\top (y-x)$ (*graph above tangent*) $\iff f(\lambda x + (1-\lambda)y) \leq \lambda f(x) + (1-\lambda) f(y)$ (*chord above graph*). Then $\nabla f(x^*) = 0 \Rightarrow$ **global** min; the set of minimizers is convex; every local min is global; no saddles or strict local maxima in the interior. Concave: mirror for max. Strict definiteness $\Rightarrow$ strict convexity, not converse (e.g. x^4 strictly convex, $H_f(0) = 0$).
- Existence: f continuous on compact $\Rightarrow$ max and min attained.

16. Lagrange & KKT

Equality constraints. $\max f(x)$ s.t. $h_j(x) = 0$: $\mathcal{L} = f - \sum \lambda_j h_j$; FOC $\nabla f = \sum \lambda_j \nabla h_j$, $h_j = 0$; λ free in sign; CQ: $\{\nabla h_j\}$ LI at x^*.

KKT — $\max f(x)$ **s.t.** $g_i(x) \geq 0$ $\mathcal{L} = f(x) + \sum_i \lambda_i g_i(x)$
(1) **Stationarity** $\nabla f(x^*) + \sum_i \lambda_i \nabla g_i(x^*) = 0$
(2) **Primal feas.** $g_i(x^*) \geq 0$
(3) **Dual feas.** $\lambda_i \geq 0$
(4) **Compl. slackness** $\lambda_i g_i(x^*) = 0$
$\lambda_i > 0 \Rightarrow$ constraint binds; slack $\Rightarrow \lambda_i = 0$.

- For a **min**: minimize $-f$, or use $\mathcal{L} = f - \sum \lambda_i g_i$ with the same signs on λ, g.
- Rewrite every constraint as $g \geq 0$ first (e.g. $w - px \geq 0$, $x \geq 0$) so the conventions hold.
- Case method: guess which constraints bind, solve, then *check* $\lambda \geq 0$ and $g \geq 0$.
- **Solving a Lagrangian — Steps:** (1) $\mathcal{L} = f(x) - \lambda^\top h(x) + \mu^\top g(x)$. (2) FOC: $\nabla_x \mathcal{L} = 0$. (3) Guess slack ($\mu = 0$) for constraints expected non-binding; solve, check $g(x) \geq 0$. (4) If infeasible, set binding $g_i(x) = 0$, re-solve treating μ_i unknown. (5) Verify compl. slack. $\mu_i g_i = 0$, dual feas. $\mu \geq 0$, primal feas. $g \geq 0, h = 0$. (6) Confirm optimum: bordered Hessian / SOC, or concavity+Slater $\Rightarrow$ KKT suff.
- **LICQ:** gradients of the binding constraints (∇h_j all j, ∇g_i for binding i) linearly independent at $x^* \Rightarrow$ KKT *necessary*. **Slater** (convex probs): $\exists \hat{x}$ with $g_i(\hat{x}) > 0$ $\forall i$, $h_j(\hat{x}) = 0$ — weaker than LICQ.
- f concave, g_i concave (feasible set convex) $\Rightarrow$ KKT also **sufficient**, global max.
- **Multiplier:** $\lambda_i = \partial V / \partial c_i$ = shadow price. **Envelope:** $dV/d\theta = \partial \mathcal{L}/\partial \theta$ at (x^*, λ^*).
- **Bordered Hessian** (n vars, m constraints): check the last $n - m$ leading principal minors of $\bar{H} = \begin{pmatrix} 0 & Dh \\ Dh^\top & H_{\mathcal{L}} \end{pmatrix}$. Max: signs alternate starting $(-1)^{m+1}$. Min: all of sign $(-1)^m$.
- **Homogeneous/homothetic:** $f(tx) = t^k f(x) \Rightarrow$ homog. degree k. Homothetic = strictly monotonic transform of a homogeneous function.
- **Log-linearization:** near a steady state, use % deviations $\hat{z} = (z - z^*)/z^*$.

17. Probability Basics

- Sample space Ω, events $\mathcal{F}$, measure $\mathbb{P}$: $\mathbb{P}(A) \geq 0$, $\mathbb{P}(\Omega) = 1$, countable additivity on disjoint sets.
- $\mathbb{P}(A^c) = 1 - \mathbb{P}(A)$; $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$; monotone: $A \subseteq B \Rightarrow \mathbb{P}(A) \leq \mathbb{P}(B)$.
- **Conditional:** $\mathbb{P}(A \mid B) = \dfrac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}$, $\mathbb{P}(B) > 0$. **Multiplication:** $\mathbb{P}(A \cap B) = \mathbb{P}(A \mid B)\mathbb{P}(B)$.
- **Independence:** $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B) \iff \mathbb{P}(A \mid B) = \mathbb{P}(A)$. Pairwise $\not\Rightarrow$ mutual.
- **Total probability:** $\{B_i\}$ a partition $\Rightarrow \mathbb{P}(A) = \sum_i \mathbb{P}(A \mid B_i)\mathbb{P}(B_i)$.
- **Bayes:** $\mathbb{P}(B_j \mid A) = \dfrac{\mathbb{P}(A \mid B_j)\mathbb{P}(B_j)}{\sum_i \mathbb{P}(A \mid B_i)\mathbb{P}(B_i)}$.
- **CDF** $F(x) = \mathbb{P}(X \leq x)$: nondecreasing, right-continuous, $F(-\infty) = 0$, $F(\infty) = 1$. Continuous case $f = F'$, $\mathbb{P}(a < X \leq b) = F(b) - F(a)$.

18. Expectation & Moments

- $\mathbb{E}[X] = \sum_x x\, p(x)$ or $\int x f(x) dx$. **LOTUS:** $\mathbb{E}[g(X)] = \int g(x) f(x) dx$ — no need for the density of $g(X)$.
- **Linearity** $\mathbb{E}[aX + bY + c] = a\mathbb{E}X + b\mathbb{E}Y + c$ *always* (no independence needed).
- $\operatorname{Var} X = \mathbb{E}[(X - \mathbb{E}X)^2] = \mathbb{E}[X^2] - (\mathbb{E}X)^2 \geq 0$; $\operatorname{Var}(aX + b) = a^2 \operatorname{Var} X$.
- **Covariance:** $\operatorname{Cov}(X,Y) = \mathbb{E}[(X - \mathbb{E}X)(Y - \mathbb{E}Y)] = \mathbb{E}[XY] - \mathbb{E}X\,\mathbb{E}Y$; $\operatorname{Cov}(X,X) = \operatorname{Var} X$; symmetric and bilinear: $\operatorname{Cov}(aX + b, cY + d) = ac \operatorname{Cov}(X,Y)$. $\operatorname{Var}(X+Y) = \operatorname{Var} X + \operatorname{Var} Y + 2\operatorname{Cov}(X,Y)$; general: $\operatorname{Var}(\sum a_i X_i) = \sum_i \sum_j a_i a_j \operatorname{Cov}(X_i, X_j)$.
- Independent $\Rightarrow \mathbb{E}[XY] = \mathbb{E}X\mathbb{E}Y \Rightarrow \operatorname{Cov} = 0$. Converse false ($X \sim N(0,1)$, $Y = X^2$).
- $\rho = \dfrac{\operatorname{Cov}(X,Y)}{\sigma_X \sigma_Y} \in [-1, 1]$ (Cauchy–Schwarz); $|\rho| = 1 \iff Y$ affine in X a.s.
- **Jensen:** g convex $\Rightarrow \mathbb{E}[g(X)] \geq g(\mathbb{E}X)$ (concave: reverse). E.g. $\mathbb{E}[X^2] \geq (\mathbb{E}X)^2$.
- **Markov:** $X \geq 0$, $a > 0 \Rightarrow \mathbb{P}(X \geq a) \leq \mathbb{E}X/a$. **Chebyshev:** $\mathbb{P}(|X - \mu| \geq k\sigma) \leq 1/k^2$.
- **MGF** $M_X(t) = \mathbb{E}[e^{tX}]$; $M^{(k)}(0) = \mathbb{E}[X^k]$; determines the distribution; independent $\Rightarrow M_{X+Y} = M_X M_Y$.
- **Sample variance:** $S_n^2 = \frac{1}{n-1}\sum (X_i - \bar{X}_n)^2$ is unbiased, $\mathbb{E}[S_n^2] = \sigma^2$ ($n-1$, not n, corrects for estimating μ by $\bar{X}_n$).

Law of Iterated Expectations (LIE)

$$\mathbb{E}[X] = \mathbb{E}\big[\mathbb{E}[X \mid Y]\big]$$

More generally $\mathbb{E}[X \mid Z] = \mathbb{E}\big[\mathbb{E}[X \mid Y, Z] \mid Z\big]$.
Key facts: $\mathbb{E}[X \mid Y]$ is a *random variable* (a function of Y); $\mathbb{E}[g(Y)X \mid Y] = g(Y)\mathbb{E}[X \mid Y]$ (*taking out what is known*); $X \perp Y \Rightarrow \mathbb{E}[X \mid Y] = \mathbb{E}X$.
Law of total variance: $\operatorname{Var} X = \mathbb{E}[\operatorname{Var}(X \mid Y)] + \operatorname{Var}(\mathbb{E}[X \mid Y])$.

- $\mathbb{E}[X \mid Y]$ is the best predictor of X given Y in mean square: it minimizes $\mathbb{E}[(X - g(Y))^2]$ over all g. Residual $X - \mathbb{E}[X \mid Y]$ is uncorrelated with any $g(Y)$ — the projection/orthogonality logic of §6.
- Discrete form: $\mathbb{E}[X] = \sum_y \mathbb{E}[X \mid Y = y]\mathbb{P}(Y = y)$.
- Normal: $X \sim N(\mu, \sigma^2) \Rightarrow aX + b \sim N(a\mu + b, a^2\sigma^2)$; sums of independent normals are normal.
- **LLN:** $\bar{X}_n \to \mu$ (in probability / a.s.). **CLT:** $\sqrt{n}(\bar{X}_n - \mu) \xrightarrow{d} N(0, \sigma^2)$.

20. Counterexample Bank

- **Partials exist, not differentiable/continuous:** $f = \frac{xy}{x^2+y^2}$, $f(0,0) = 0$: $f_x = f_y = 0$ at 0 but $f \to \frac{1}{2}$ along $y = x$. (Polar: $f = \frac{1}{2}\sin 2\theta$, constant on rays.)
- **All directional derivatives, still not differentiable:** $f = \frac{x^2 y}{x^2+y^2}$, $f(0,0) = 0$; $D_v f(0,0) = \frac{a^2 b}{a^2+b^2}$ is *not linear* in v.
- **Continuous image of closed need not be closed:** $\frac{1}{1+x^2}$ on $\mathbb{R}$ gives $(0,1]$.
- **Continuous image of bounded need not be bounded:** $1/x$ on $(0,1)$.
- **No max:** x on $(0,1)$ (bounded, not closed); x on $[0,\infty)$ (closed, not bounded); $1/x$ on $(0,1]$ with $f(0) = 0$ (compact, discontinuous).
- **Continuous bijection, discontinuous inverse:** $t \mapsto (\cos t, \sin t)$ on $[0, 2\pi)$ — domain not compact.
- Df **singular but 1-1:** x^3. Df **invertible, not globally 1-1:** $e^x(\cos y, \sin y)$.
- **Not diagonalizable:** $\left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$ over $\mathbb{C}^2$.
- **Uncorrelated but dependent:** $X \sim N(0,1)$, $Y = X^2$.

21. Diagnose the Statement

claim	
differentiable $\Rightarrow$ continuous	T
all partials exist $\Rightarrow$ differentiable	F
all directional derivatives exist $\Rightarrow$ diff.	F
partials near x + continuous at $x \Rightarrow$ diff.	T
continuous image of compact is compact	T
continuous image of closed is closed	F
closed subset of compact is compact	T
$Df(x)$ invertible $\Rightarrow$ globally 1-1	F
$Df(x)$ singular $\Rightarrow$ no *diff.* local inverse	T
$Df(x)$ singular $\Rightarrow$ not locally 1-1	F
convex U, $Df \equiv 0 \Rightarrow f$ constant	T
convex U, $H_f \equiv 0 \Rightarrow f$ affine	T
$\operatorname{rank}(A^T A) = \operatorname{rank}(A)$	T
$f \in C^2$ strictly convex $\implies \nabla^2 f(x) \succ 0$	F
$X_n \xrightarrow{\text{a.s.}} X \implies X_n \xrightarrow{p} X$	T

22. Proof Moves

- Neighborhood property fails $\Rightarrow$ **build a sequence** with $\varepsilon = 1/k$, extract, converge.
- Set open/closed $\Rightarrow$ write it as f^{-1}(open/closed) for continuous f. Never form an image.
- **Preimage skeleton:** $a \in f^{-1}(V)$ means $f(a) \in V$; openness of V gives ε; continuity gives δ; then $x \in B_\delta(a) \Rightarrow f(x) \in B_\varepsilon(f(a)) \subseteq V \Rightarrow x \in f^{-1}(V)$. Conclude $B_\delta(a) \subseteq f^{-1}(V)$.
- $x_k \not\to x \Rightarrow \exists \varepsilon > 0$ and a subsequence bounded away; compactness gives a *further* convergent subsequence; contradict via uniqueness of limits + injectivity.
- Bounding an inf you cannot evaluate: fix an arbitrary element, derive the inequality for all of them, then take inf on the side that does not depend on it.
- $A = B$ for linear maps $\Rightarrow$ show $(A - B)v = 0$ for all v.
- Multivariate statement $\Rightarrow$ restrict to a line, use MVT or 1-D Taylor.
- $a = b \Rightarrow$ prove $\leq$ and $\geq$, or $|a - b| < \varepsilon$ for every $\varepsilon > 0$.
- Showing something is *not* linear: exhibit v_1, v_2 with $T(v_1 + v_2) \neq T(v_1) + T(v_2)$.